

Advanced Computer Networks – PCS223

Assignment 5

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Subgroup – CS4

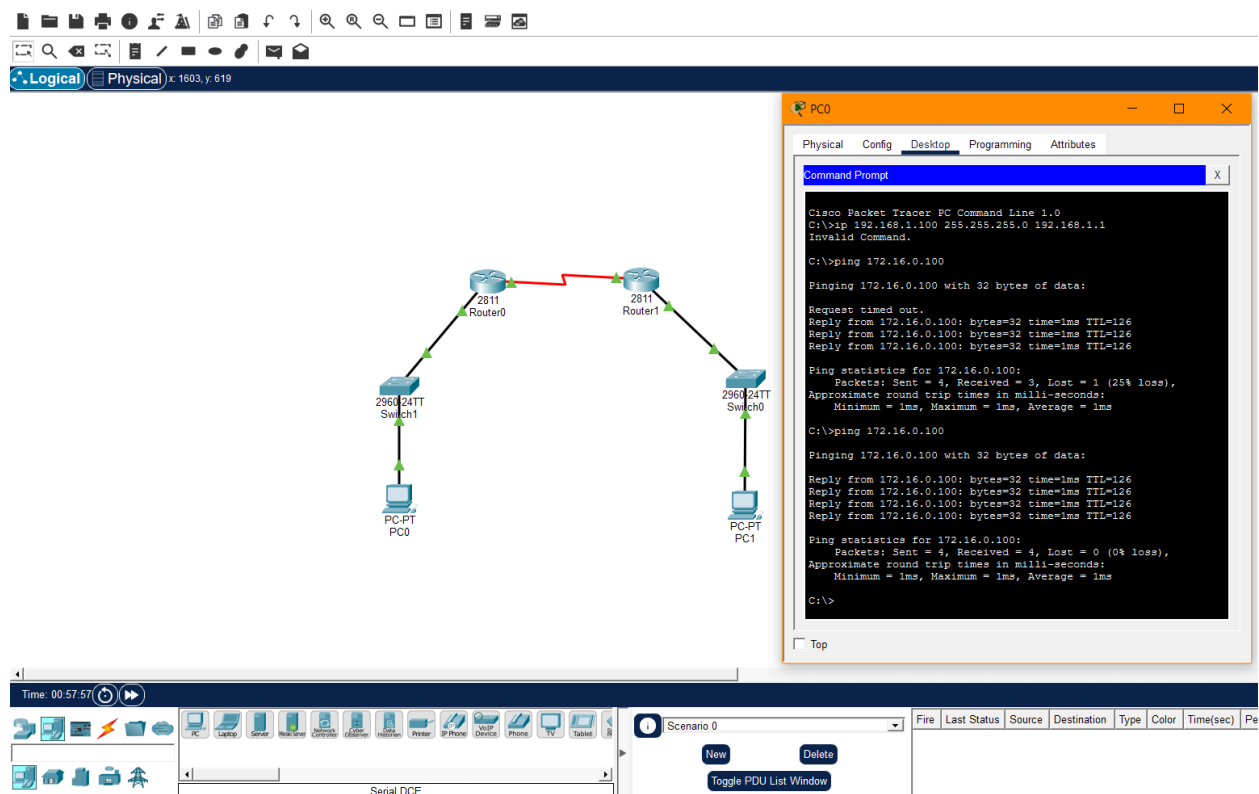
Roll No. – 8025320103

Q.1 Objective: Configure Static Route in Cisco packet tracer

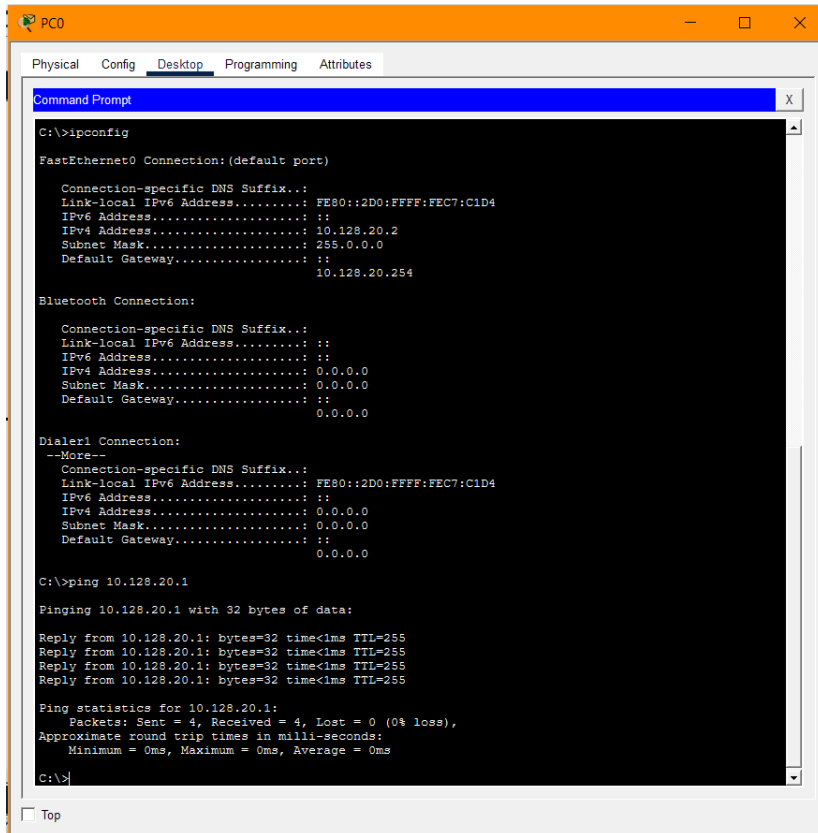
- Static route tells the device exactly where to send traffic, no matter what.
- Static route is often used when your network has only a few routers or there is only one route from a source to a destination.

Scenario

- Suppose that your company has 2 branches located in Tehran and Shiraz.
- As the administrator of the network, you are tasked to connect them so that employees in the two LANs can communicate with each other.
- After careful consideration you decided to connect them via static route.



Q.2 Suppose you want to create 3 subnets of a network having network address 198.168.1.0. In another network with network address 220.168.2.0, you want to create 2 subnets. For this scenario configure the end devices and routers accordingly. Connect two different networks (both are subnetted) through static routing and show the message transmission among hosts of the two networks.



```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:FFFF:FEC7:C1D4
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 10.128.20.2
    Subnet Mask . . . . .: 255.0.0.0
    Default Gateway . . . . .: 10.128.20.254

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .:
    IPv6 Address . . . . .:
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .:

Dialer1 Connection:
--More--
    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:FFFF:FEC7:C1D4
    IPv6 Address . . . . .:
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .:

C:\>ping 10.128.20.1

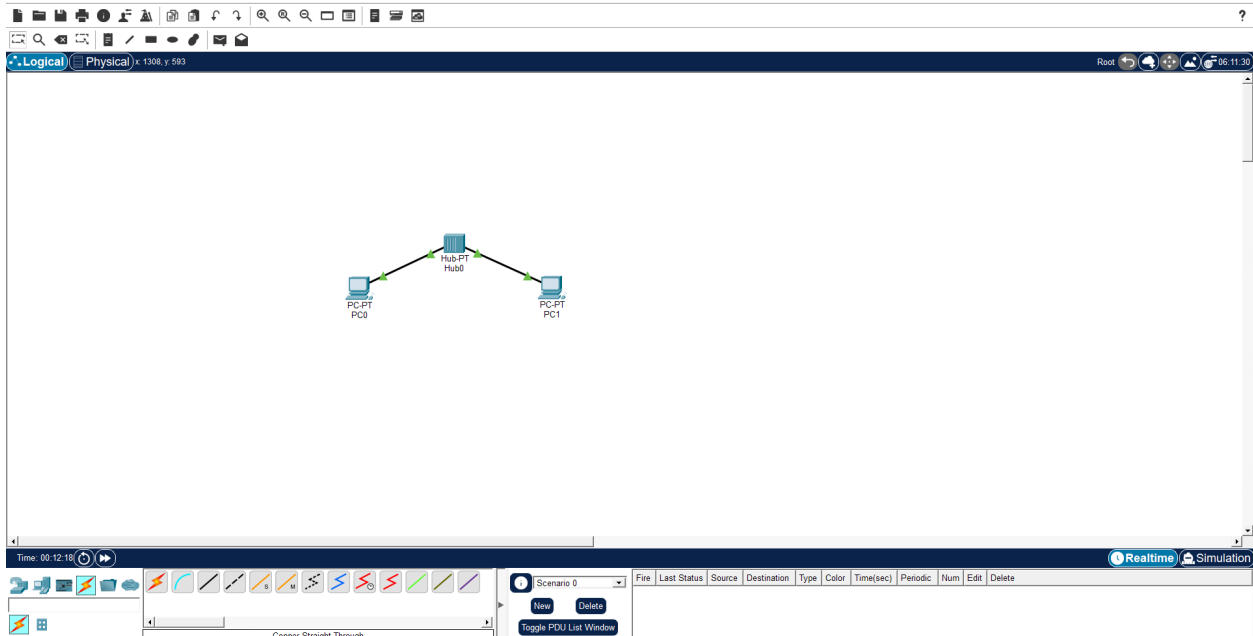
Pinging 10.128.20.1 with 32 bytes of data:

Reply from 10.128.20.1: bytes=32 time<1ms TTL=255
Reply from 10.128.20.1: bytes=32 time<1ms TTL=255
Reply from 10.128.20.1: bytes=32 time<1ms TTL=255
Reply from 10.128.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.128.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

1. Create a network of 2 PCs as below.



PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

☐ Top

2. Connect 2 PCs with a Switch below.

Logical Physical x:950, y:279

Time: 00:00:51

Scenario 0

New Delete

Toggle PDU List Window

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\> ping 192.168.1.2

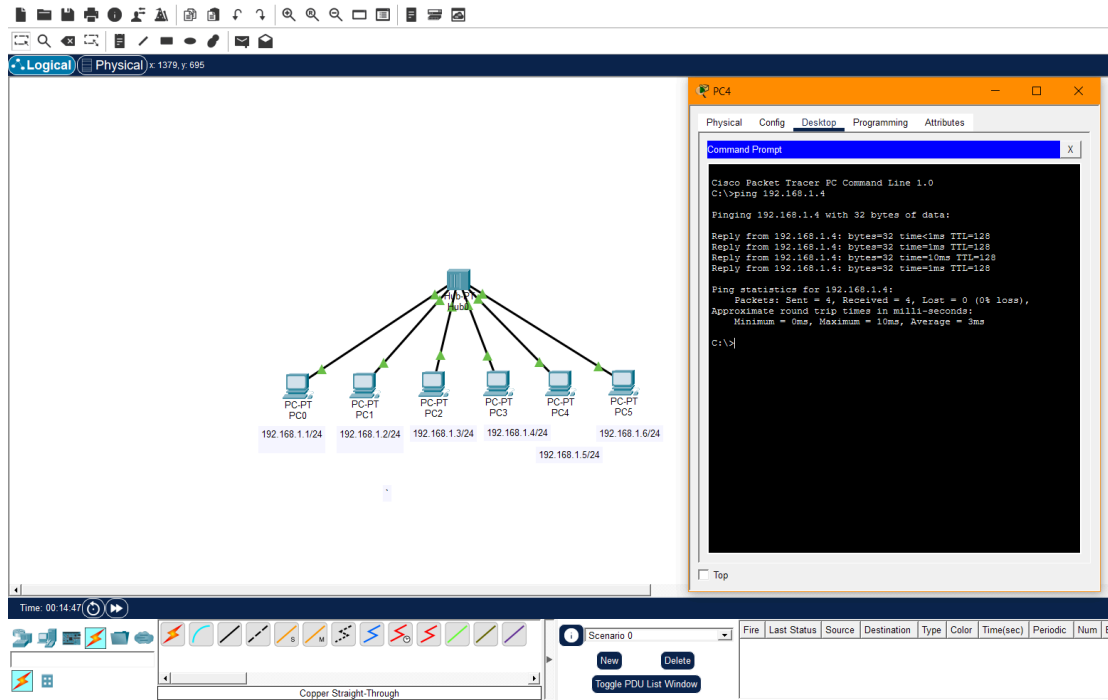
Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

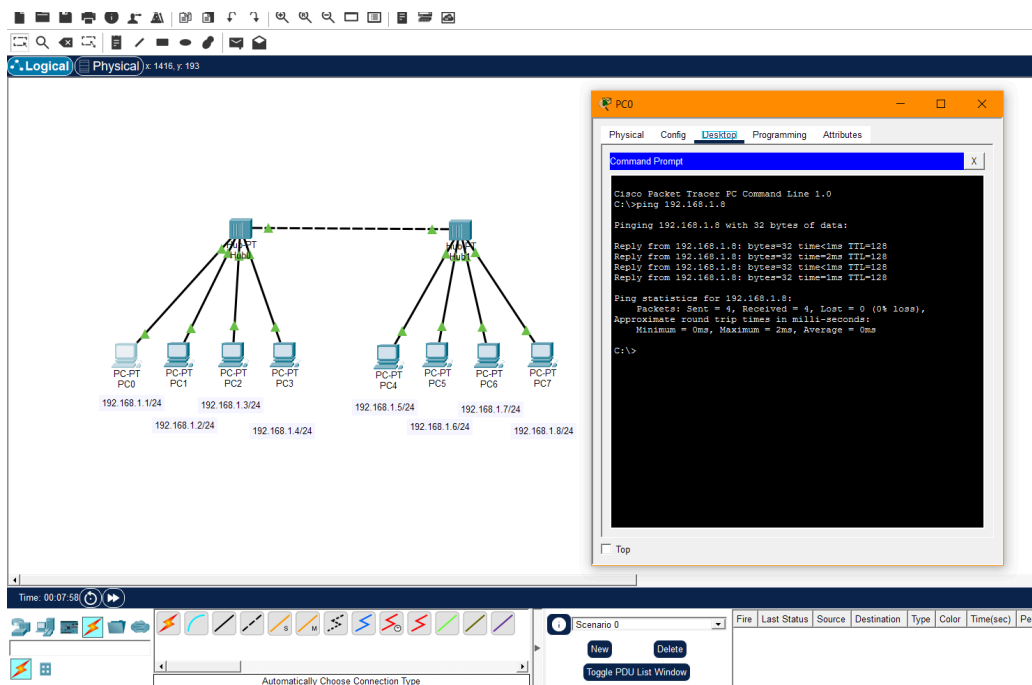
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```

Top

3. Create BUS topologies as below.

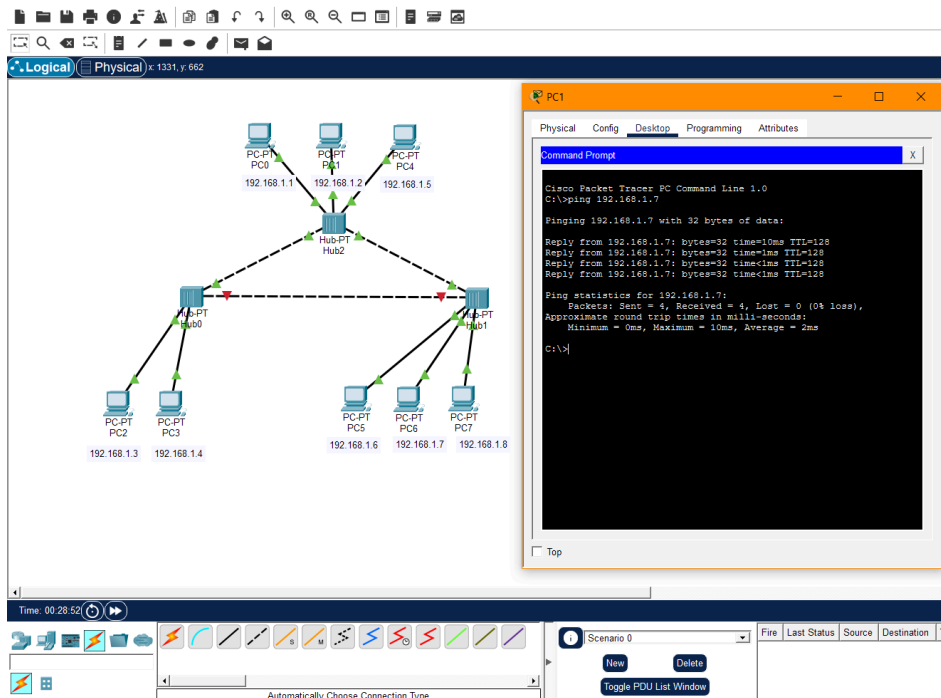


(a)

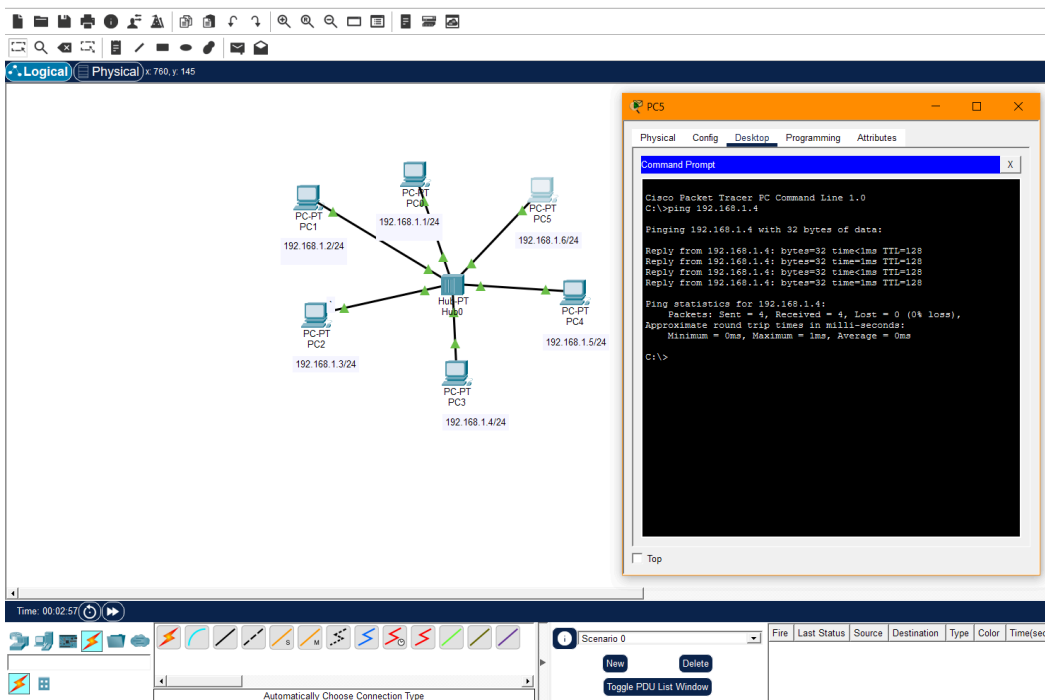


(b)

4. Create the following ring topology



5. Create the following ring topology



6. Create the following ring topology

LogicalPhysicalx 1370, y 626

PC3

PhysicalConfigDesktopProgrammingAttributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.7

Pinging 192.168.1.7 with 32 bytes of data:
Reply from 192.168.1.7: bytes=32 time=10ms TTL=128
Reply from 192.168.1.7: bytes=32 time=1ms TTL=128
Reply from 192.168.1.7: bytes=32 time=10ms TTL=128
Reply from 192.168.1.7: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 5ms

C:\>
```

Top

Time: 00:03:48

Copper Straight-Through

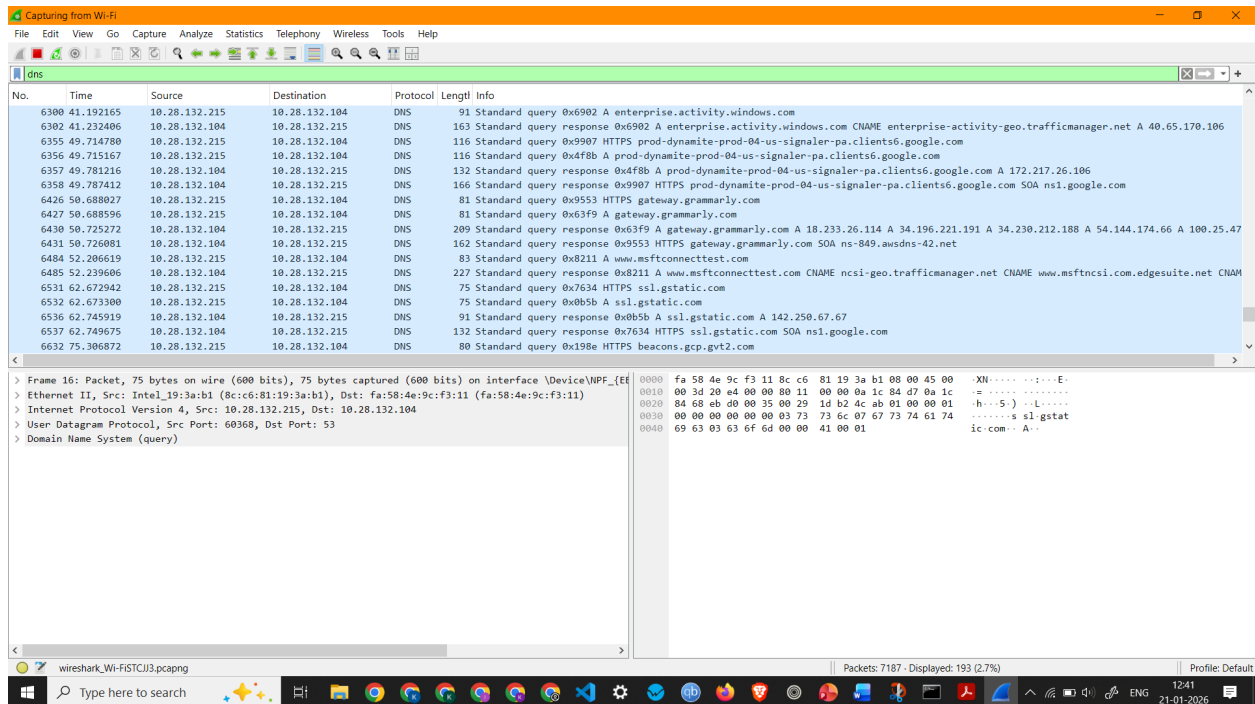
Scenario 0

NewDelete

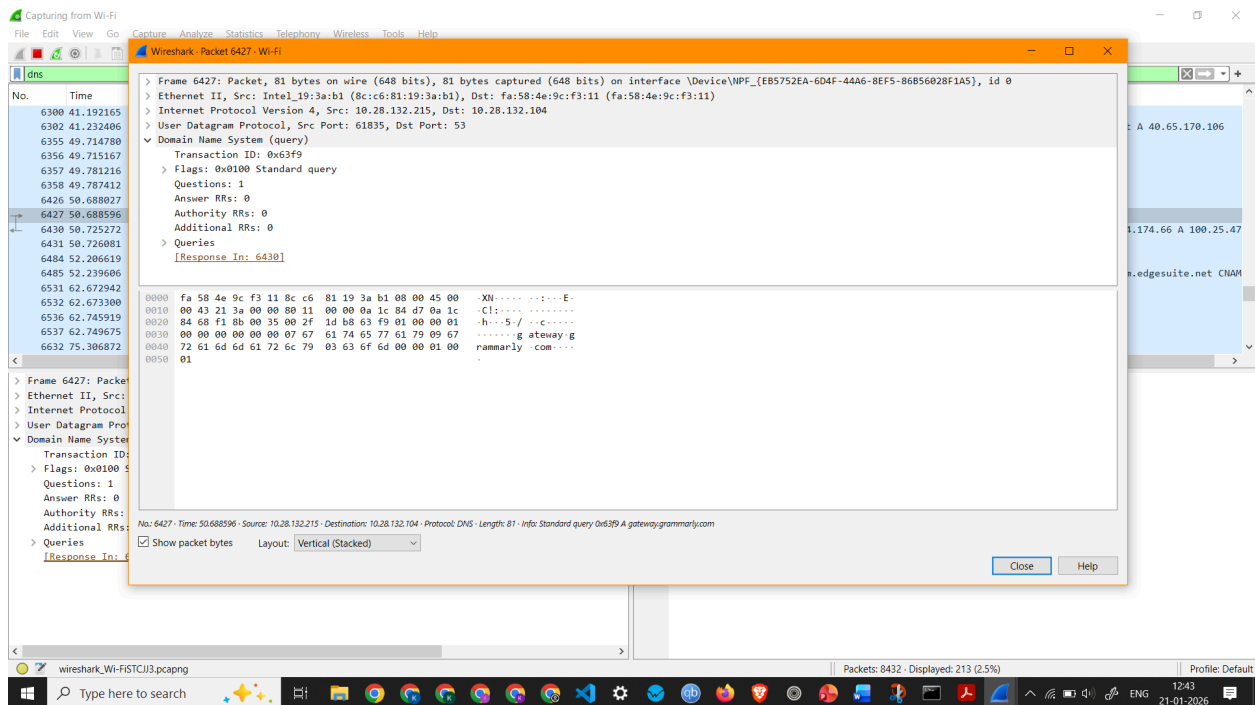
Toggle PDU List Window

FireLast StatusSourceDestination

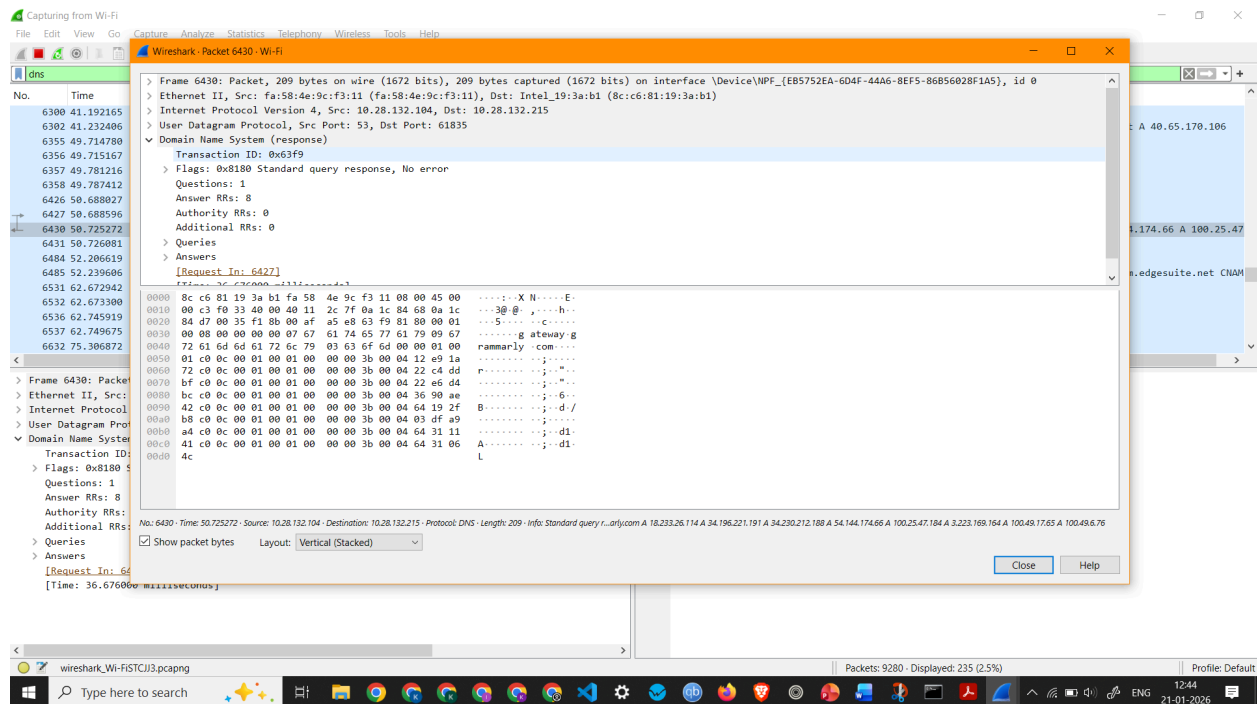
4. To capture and analyze DNS query and response packets



Screenshot showing DNS Packets Capture



Screenshot showing details of a DNS Packet 6427 (Standard query request)



Screenshot showing details of a DNS Packet 6430 (Standard query response)

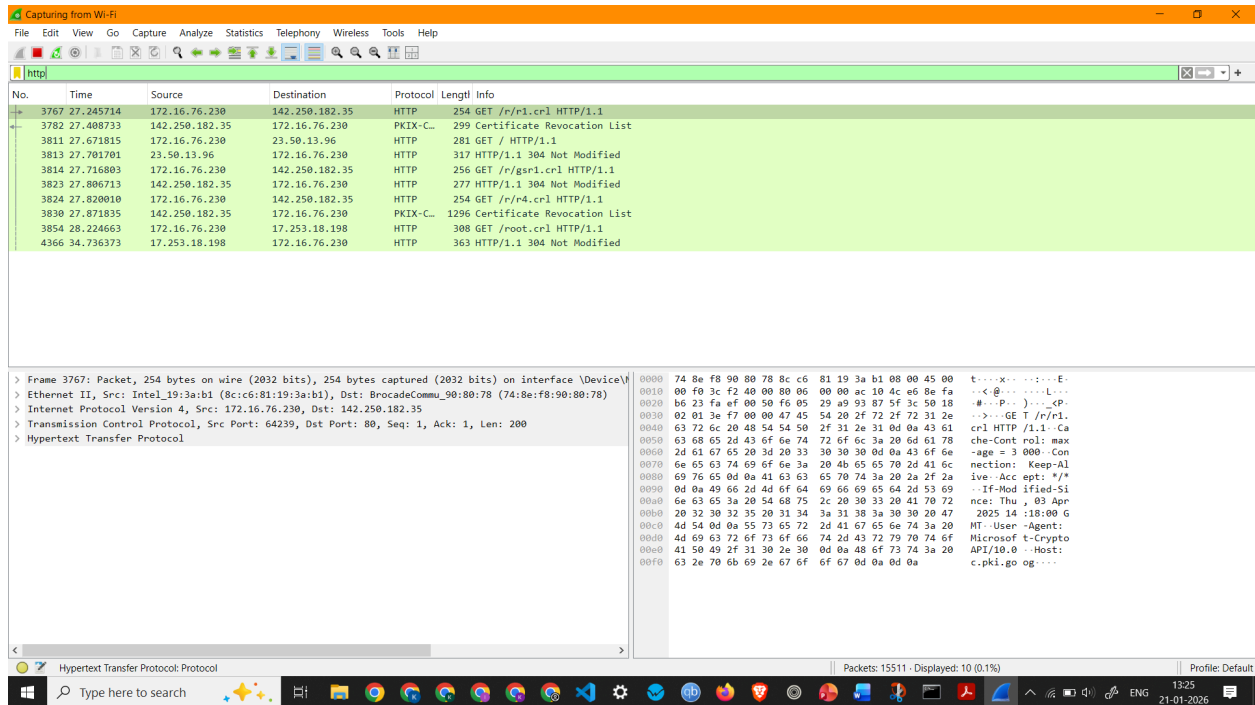
Observation: DNS Packet 6427 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 10.28.132.215
- Destination IP Address: 10.28.132.104
- Packet Length: 81 Bytes
- Queries - gateway.grammarly.com: type A, class IN

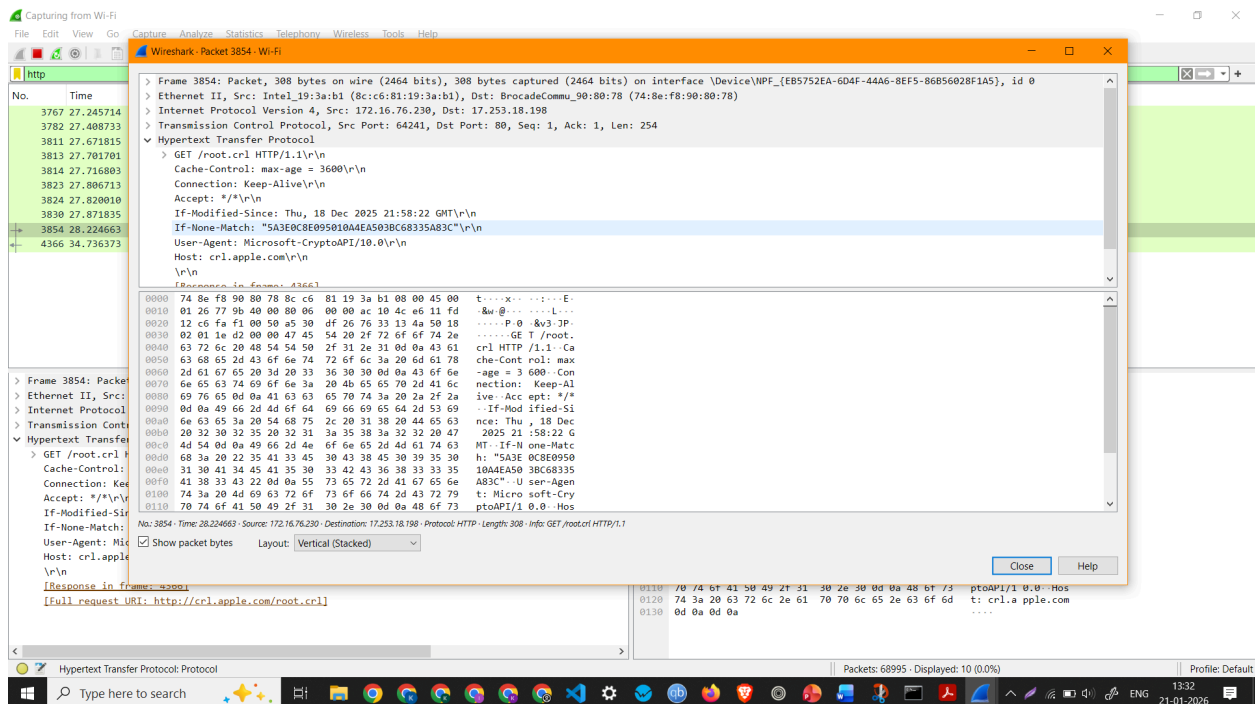
Observation: DNS Packet 6430 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 10.28.132.104
- Destination IP Address: 10.28.132.215
- Packet Length: 209 Bytes

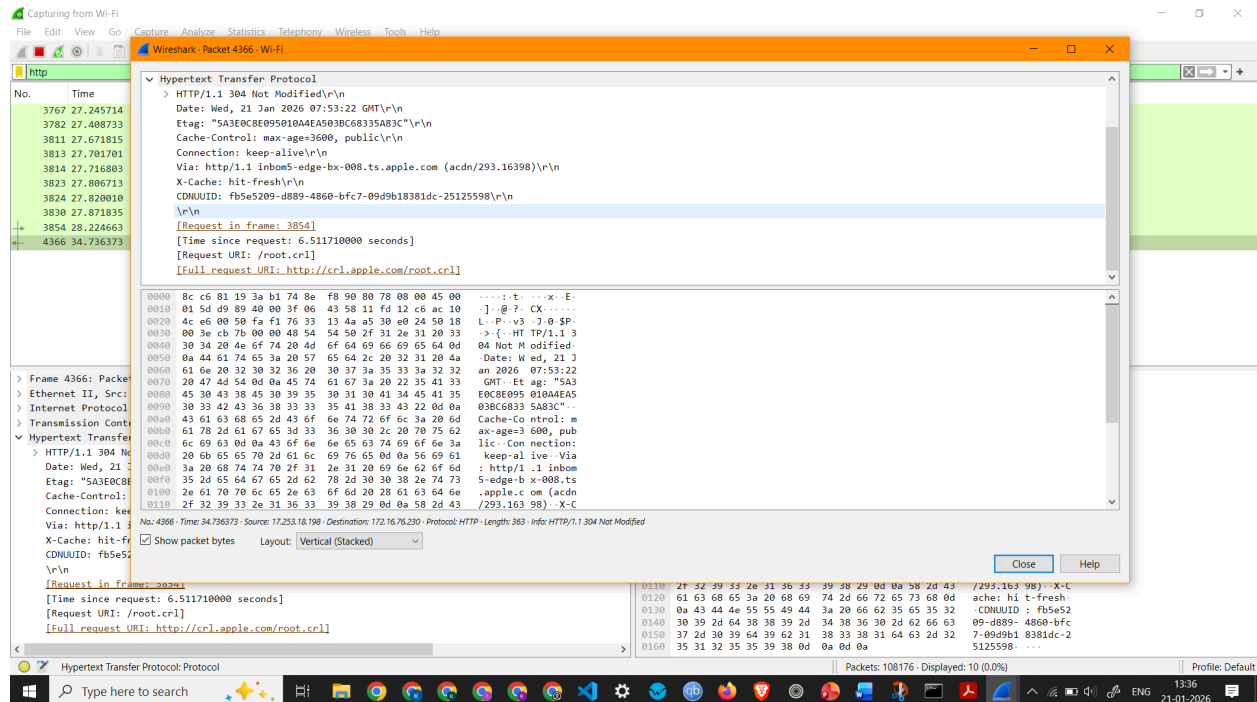
5. To capture and analyze HTTP packets



Screenshot showing HTTP Packets Capture



Screenshot showing details of a HTTP Packet (HTTP Request)



Screenshot showing details of a HTTP Packet (HTTP Response)

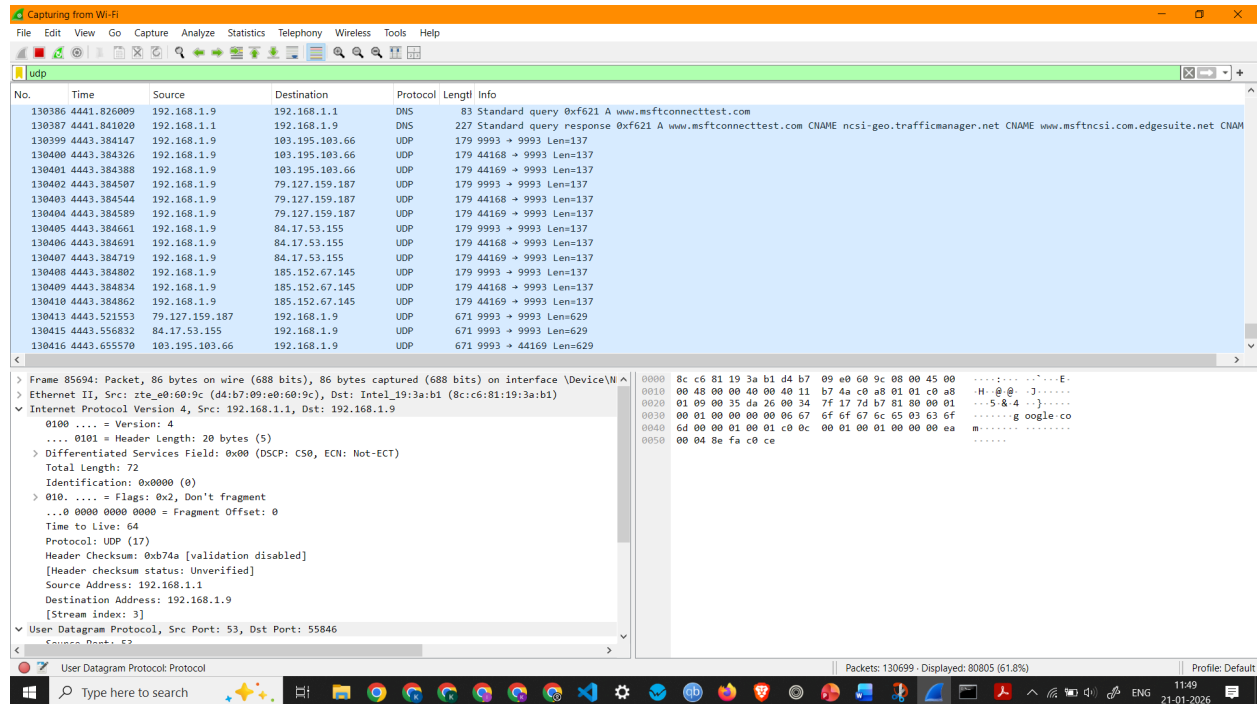
Observation: HTTP Request 3854 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 172.16.76.230
- Destination IP Address: 17.253.18.198
- Packet Length: 308 Bytes

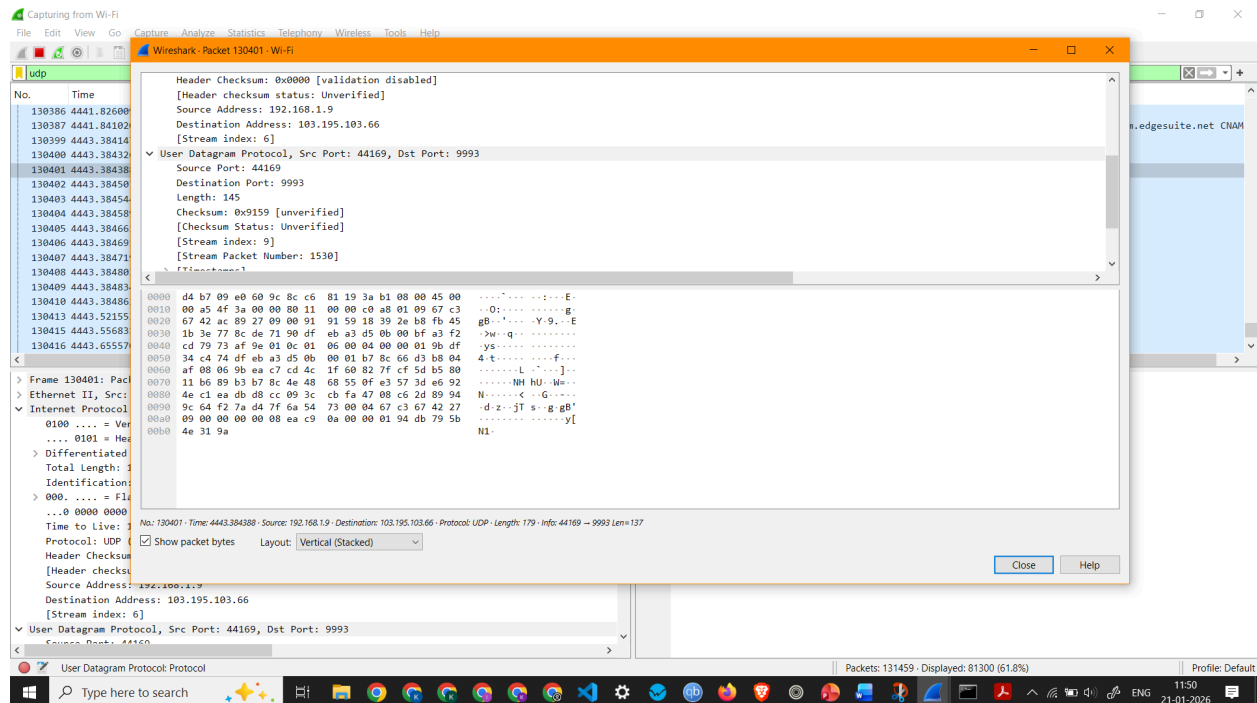
Observation: HTTP Response 4366 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 17.253.18.198
- Destination IP Address: 172.16.76.230
- Packet Length: 363 Bytes

3. To capture and analyze UDP packets



Screenshot showing UDP Packets Capture

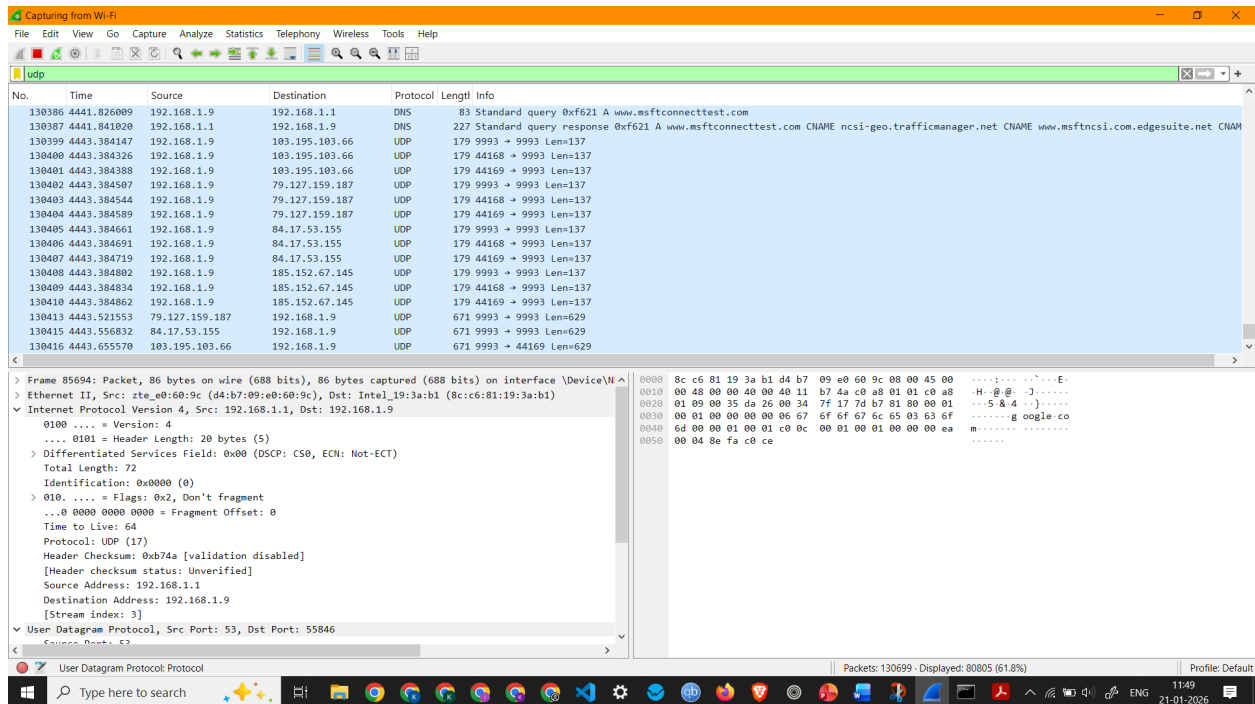


Screenshot showing details of a UDP Packet

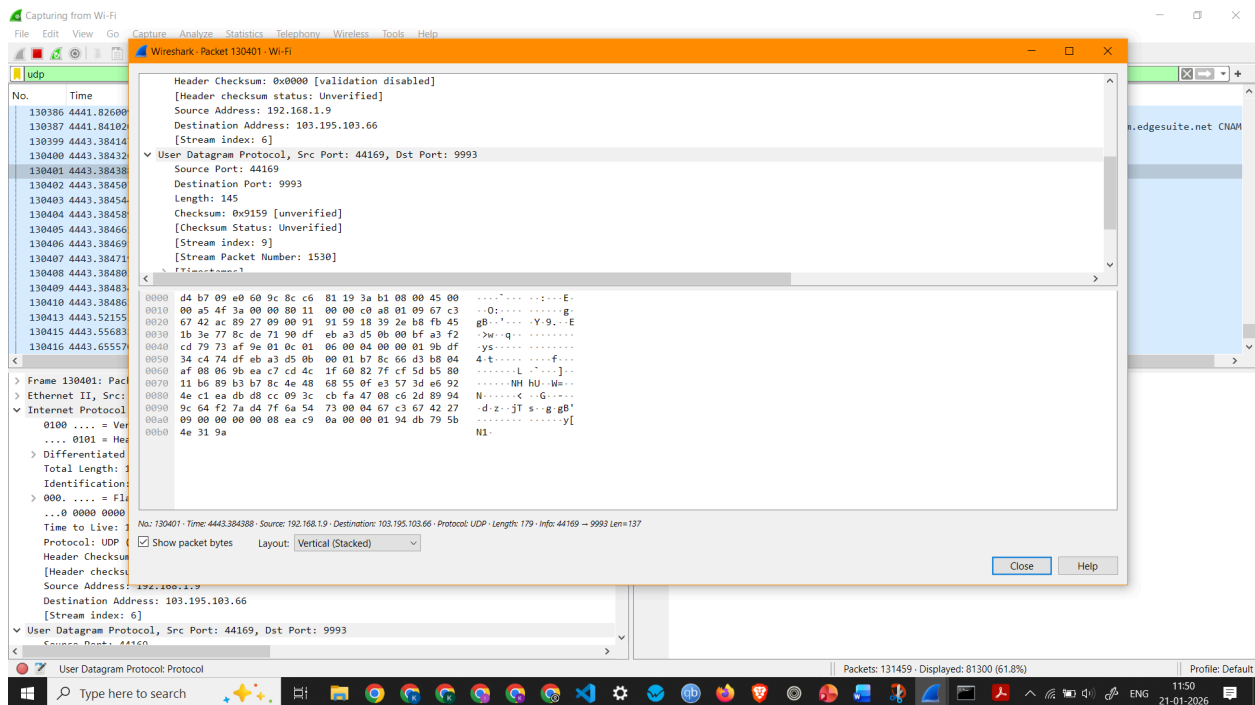
Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes

3. To capture and analyze UDP packets



Screenshot showing UDP Packets Capture

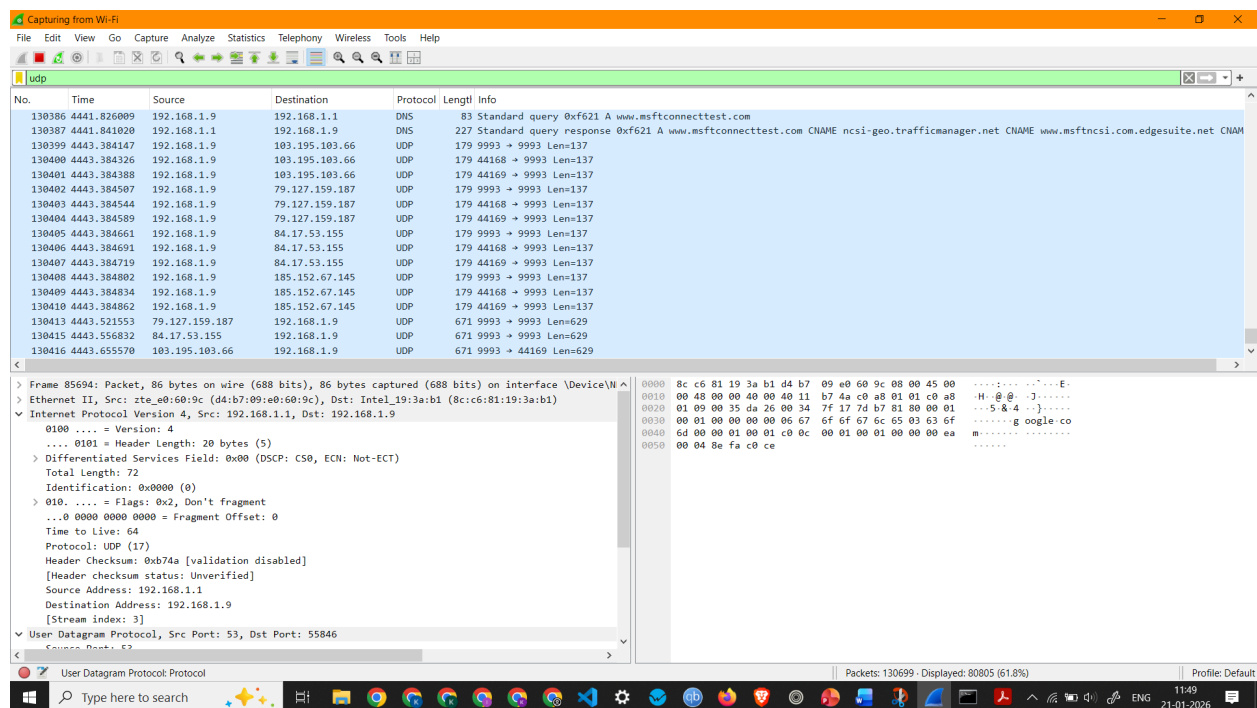


Screenshot showing details of a UDP Packet

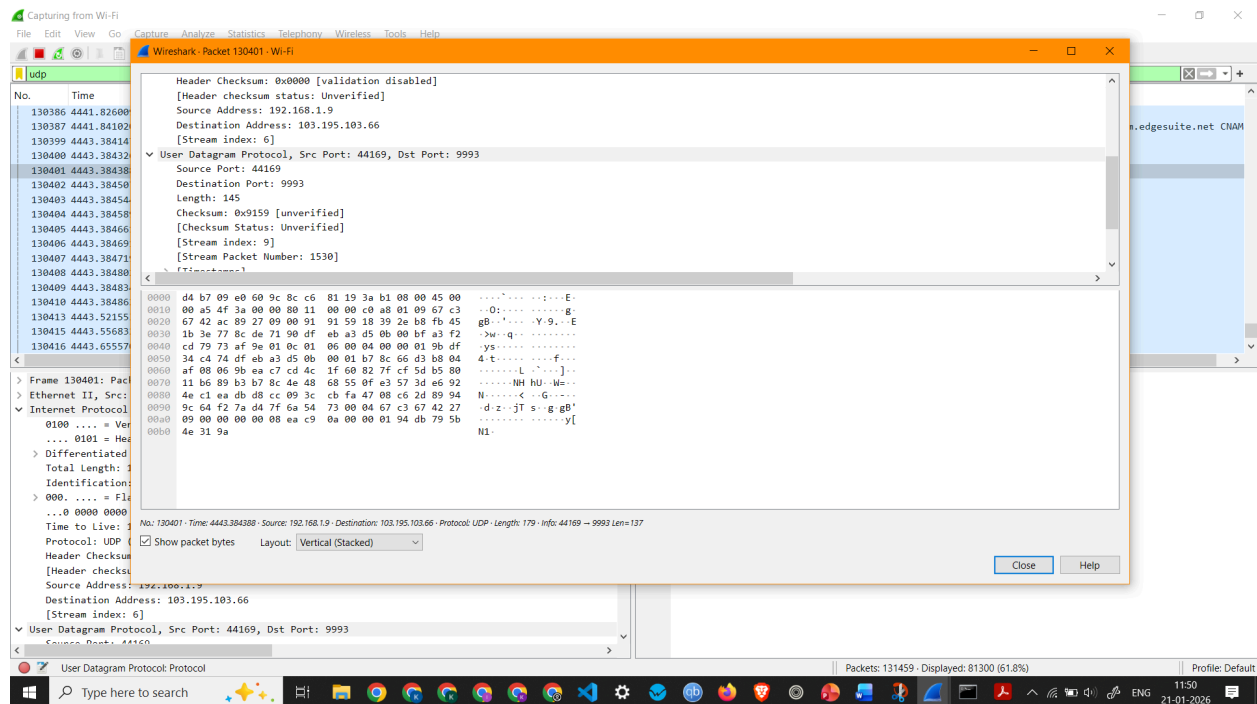
Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes

3. To capture and analyze UDP packets



Screenshot showing UDP Packets Capture

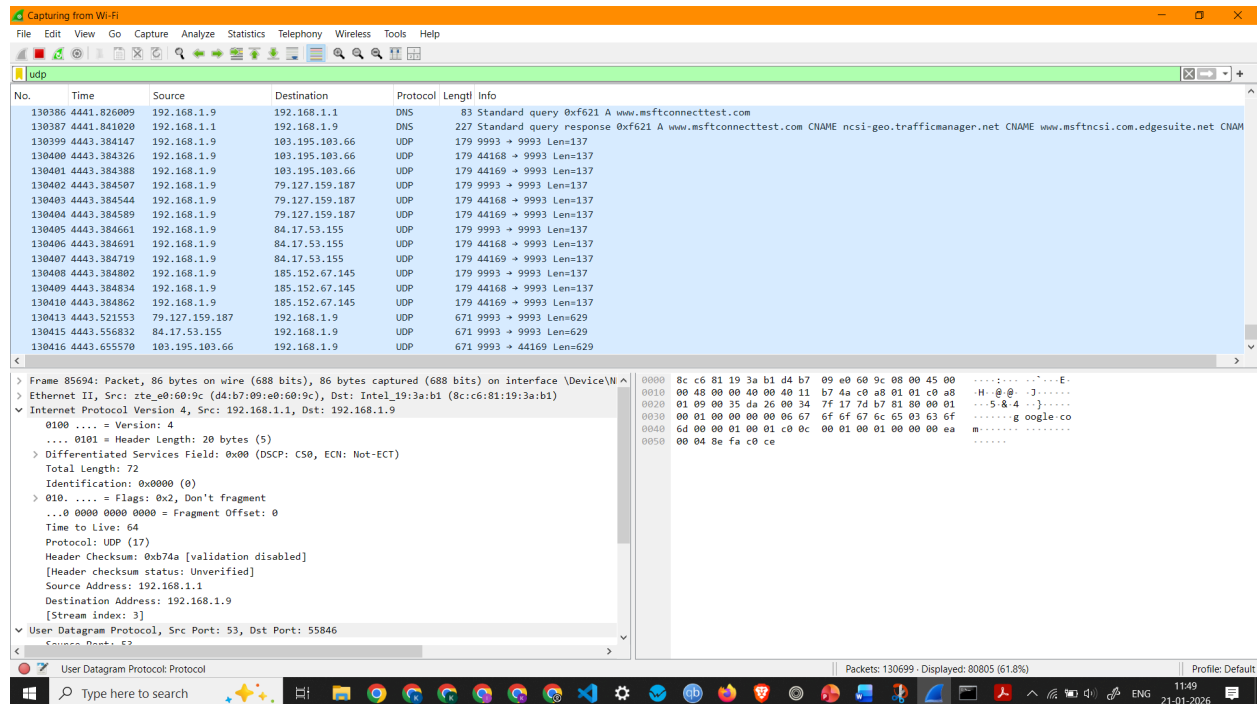


Screenshot showing details of a UDP Packet

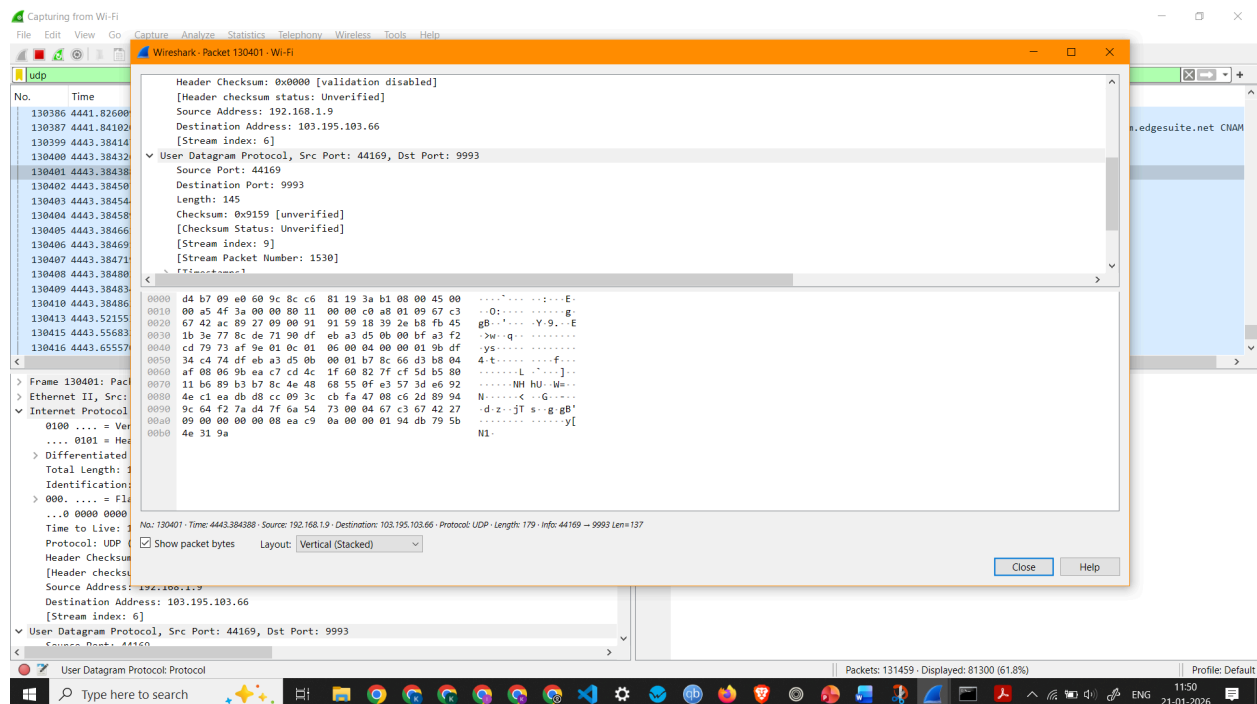
Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes

3. To capture and analyze UDP packets



Screenshot showing UDP Packets Capture



Screenshot showing details of a UDP Packet

Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes